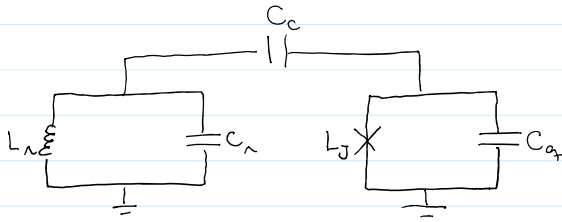
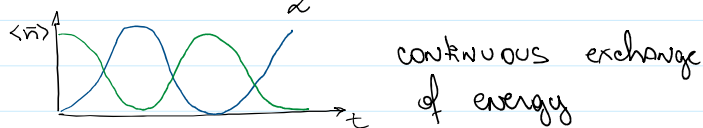


Last Lecture



Capacitive coupling

$$H_{JC} = \hbar \omega_L a^\dagger a + \frac{\hbar \omega_J}{2} \sigma_z + \hbar g (a \sigma_+ + a^\dagger \sigma_-)$$



$$[H_{JC}, N] = 0 \rightarrow \text{total excitation number is conserved}$$

Today: How do we measure each state?

We could easily measure $a^\dagger a$ or σ_z , but

$$[H_{JC}, a^\dagger a] \neq 0 \quad [H_{JC}, \sigma_z] \neq 0$$

then, individually, they are not constant. A measurement of these operators, in this regime, would alter the state of the system. The measurement strongly modifies the state of interest.

We look for a situation where we measure an operator R such that

$$[H, R] = 0.$$

The measurement does not modify the state: Quantum Non-demolition measurement (QND).

Useful tool:

Rotating Frame Transformation

We can transform the Hamiltonian with unitary transformations

$$|\tilde{\Psi}(t)\rangle = U(t)|\Psi(t)\rangle, \quad UU^\dagger = \mathbb{1}$$

Let's write a new Schrödinger equation

$$\frac{d}{dt} |\tilde{\Psi}(t)\rangle = \dot{U}|\Psi\rangle + U \frac{d}{dt} |\Psi\rangle$$

$\underbrace{\frac{d}{dt}}_{\text{original S.eq.}} \quad i\hbar \frac{d}{dt} |\Psi\rangle = H|\Psi\rangle$

Multiplying by $i\hbar$

$$i\hbar \frac{d}{dt} |\tilde{\Psi}(t)\rangle = i\hbar \dot{U}|\Psi(t)\rangle + UH|\Psi(t)\rangle \quad \text{but} \quad |\Psi(t)\rangle = U^\dagger |\tilde{\Psi}(t)\rangle$$

$$\rightarrow i\hbar \frac{d}{dt} |\tilde{\Psi}(t)\rangle = \underbrace{(i\hbar \dot{U}U^\dagger + UH U^\dagger)}_{\tilde{H}} |\tilde{\Psi}(t)\rangle$$

Therefore, we have for the transformed system

$$\begin{aligned} \tilde{H} &= UH U^\dagger + i\hbar \dot{U}U^\dagger \\ &\quad \downarrow \text{using } UU^\dagger = \mathbb{1} \rightarrow \dot{U}U^\dagger + U\dot{U}^\dagger = 0 \\ \tilde{H} &= UH U^\dagger - i\hbar U\dot{U}^\dagger \quad (1) \end{aligned}$$

This is useful in many situations, such as to get rid of trivial fast oscillations. 

Let's modify our H_{JC} to a frame rotating at the resonator frequency

$$U(t) = e^{i\omega_n t (a^\dagger a + \frac{\sigma_z}{2})}$$

Using (1), we solve the terms

• time derivative

$$U\dot{U}^\dagger = -i\omega_n \left(a^\dagger a + \frac{\sigma_z}{2} \right)$$

• Free terms

$$U a a^\dagger U^\dagger = a^\dagger a \quad ([U, a^\dagger a] = 0)$$

$$U \sigma_z U^\dagger = \sigma_z \quad ([U, \sigma_z] = 0)$$

• Then, free terms + time derivative term $\rightarrow$

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$$U \left(\hbar \omega_n a^\dagger a + \frac{\hbar \omega_q \sigma_z}{2} \right) U^\dagger - i \hbar \left[-i \omega_n (a^\dagger a + \frac{\sigma_z}{2}) \right]$$

$$= \hbar \omega_n a^\dagger a + \frac{\hbar \omega_q \sigma_z}{2} - \hbar \omega_n \left(a^\dagger a + \frac{\sigma_z}{2} \right) = \hbar \frac{(\omega_q - \omega_n)}{2} \sigma_z$$

$$= \frac{\hbar \Delta}{2} \sigma_z$$

with $\Delta = \omega_q - \omega_n$

• Now, the coupling terms

$$U a U^\dagger = e^{i \omega_n t a^\dagger a} \cdot a \cdot e^{-i \omega_n t a^\dagger a}$$

To solve this, we are going to use the Baker-Campbell-Hausdorff formula (BCH)

$$e^A B e^{-A} = B + [A, B] + \frac{1}{2!} [A, [A, B]] + \dots$$

So

$$U a U^\dagger = a + \underbrace{[i \omega_n a^\dagger a t, a]}_{-i \omega_n t a} + \frac{1}{2!} \underbrace{[i \omega_n a^\dagger a t, (-i \omega_n t a)]}_{\frac{1}{2!} (-i \omega_n t)^2 a} + \frac{1}{3!} \underbrace{[i \omega_n a^\dagger a t, (-i \omega_n t)^2 a]}_{\frac{1}{3!} (-i \omega_n t)^3 a} + \dots$$

$$\Rightarrow U a U^\dagger = \sum_n \frac{(-i \omega_n t)^n}{n!} a = e^{-i \omega_n t} a$$

and

$$U a^\dagger U^\dagger = e^{i \omega_n t} a^\dagger$$

Also

$$U \sigma_+ U^\dagger = e^{i \omega_n \sigma_z t / 2} \sigma_+ e^{i \omega_n \sigma_z t / 2} = \sigma_+ e^{i \omega_n t}$$

$$U \sigma_- U^\dagger = \sigma_- e^{-i \omega_n t}$$

Plugging it all together

$$\tilde{H}_{JC} = \underbrace{\frac{\hbar \Delta}{2} \sigma_z}_{\tilde{H}_0} + \underbrace{\hbar g (a \sigma_+ + a^\dagger \sigma_-)}_{\tilde{V}}$$

Exercise: Use the rotating frame transformation

$$U(t) = \exp\left[i\omega_q t a^\dagger a + i\frac{\omega_q}{2} t \sigma_z\right] \rightarrow \text{free terms cancel}$$

What is the new Hamiltonian?

$$a\sigma_+ e^{-i\omega_q t} \cdot e^{i\omega_q t} = a\sigma_+ e^{i\Delta t} \quad \Delta = \omega_q - \omega_a$$

$$\rightarrow H_{JC} = \hbar g (a\sigma_+ e^{i\Delta t} + a^\dagger \sigma_- e^{-i\Delta t})$$

Dispersive Regime

Let's now assume that, instead of a resonant interaction, we have $\Delta \gg g\sqrt{n+1}$. If this is the case, the off-diagonal terms are small and we can treat them perturbatively. This is called a dispersive regime.

To eliminate the off-diagonal terms, we will use the Schrieffer-Wolff transformation. It consists of a near-identity unitary change of basis. You use it when you have a Hamiltonian $H = H_0 + V$ with diagonal H_0 and V weakly coupling different eigenspaces.

SW-Method

- Seek an Anti-Hermitian generator $S = -S^\dagger$ such that

$$[H_0, S] = V$$

- Transform H as

$$H_{\text{eff}} = e^S H e^{-S} \stackrel{\text{ BCH }}{=} \underbrace{H + [S, H]}_{H_0 + [S, V]} + \frac{1}{2} [S, [S, H]] + \dots$$

So, let's do this with H_{JC} . We want $[H_0, S] = V$

ansatz: $S = \lambda (a\sigma_+ - a^\dagger \sigma_-)$

$$[\tilde{H}_0, S] = \left[\frac{\hbar \Delta}{2} \sigma_z, \lambda a\sigma_+ - \lambda a^\dagger \sigma_- \right] = \frac{\hbar \Delta}{2} \cdot \lambda \left(a \underbrace{[\sigma_z, \sigma_+]}_{2\sigma_+} - a^\dagger \underbrace{[\sigma_z, \sigma_-]}_{-2\sigma_-} \right)$$

$$= \hbar \Delta \lambda (a\sigma_+ + a^\dagger \sigma_-)$$

Setting this equal to $\tilde{V}$

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$$\hbar g (a\sigma_+ + a^\dagger\sigma_-) = \hbar\Delta\lambda (a\sigma_+ + a^\dagger\sigma_-) \Rightarrow \lambda = \frac{g}{\Delta}$$

Now, $\tilde{H}_{\text{eff}}$

$$\begin{aligned}\tilde{H}_{\text{eff}} &= e^S \tilde{H} e^{-S} = \tilde{H} + [S, \tilde{H}] + \frac{1}{2} [S, [S, \tilde{H}]] + \dots \\ &= \frac{\hbar\Delta\sigma_z}{2} + \hbar g (a\sigma_+ + a^\dagger\sigma_-) - \hbar g (a\sigma_+ + a^\dagger\sigma_-) + [S, \tilde{V}] \\ &\quad + \frac{1}{2} [S, [S, \tilde{H}_0]] + \frac{1}{2} [S, [S, \tilde{V}]] + \dots\end{aligned}$$

$$\tilde{H}_{\text{eff}} = \frac{\hbar\Delta\sigma_z}{2} + \frac{1}{2} [S, \tilde{V}] + O(\lambda^3)$$

we need to compute $\frac{1}{2}[S, \tilde{V}]$

$$\begin{aligned}\frac{1}{2}[S, \tilde{V}] &= \frac{\hbar g^2}{2\Delta} [a\sigma_+ - a^\dagger\sigma_-, a\sigma_+ + a^\dagger\sigma_-] \\ &= \frac{\hbar g^2}{2\Delta} ([a\sigma_+, a^\dagger\sigma_-] - [a^\dagger\sigma_-, a\sigma_+]) \\ &= \frac{\hbar g^2}{\Delta} [a\sigma_+, a^\dagger\sigma_-] = \frac{\hbar g^2}{\Delta} (aa^\dagger\sigma_- - a^\dagger a\sigma_+)\end{aligned}$$

$$\sigma_+\sigma_- = \frac{\mathbb{1} + \sigma_z}{2} \quad \sigma_-\sigma_+ = \frac{\mathbb{1} - \sigma_z}{2} \quad aa^\dagger = a^\dagger a + \mathbb{1}$$

$$\begin{aligned}\Rightarrow \frac{1}{2}[S, \tilde{V}] &= \frac{\hbar g^2}{\Delta} \left\{ (a^\dagger a + \mathbb{1}) \left(\frac{\mathbb{1} + \sigma_z}{2} \right) - a^\dagger a \left(\frac{\mathbb{1} - \sigma_z}{2} \right) \right\} \\ &= \frac{\hbar g^2}{\Delta} \left(\cancel{\frac{a^\dagger a}{2}} + \frac{a^\dagger a \sigma_z}{2} + \frac{\mathbb{1}}{2} + \frac{\sigma_z}{2} - \cancel{\frac{a^\dagger a}{2}} + \frac{a^\dagger a \sigma_z}{2} \right) \\ &= \frac{\hbar g^2}{\Delta} \left(a^\dagger a \sigma_z + \frac{\sigma_z + \mathbb{1}}{2} \right)\end{aligned}$$

Therefore, the total $\tilde{H}_{\text{eff}}$, in the rotating frame is

$$\tilde{H}_{\text{eff}} \approx \frac{\hbar\Delta}{2} \sigma_z + \frac{\hbar g^2}{\Delta} a^\dagger a \sigma_z + \frac{\hbar g^2}{2\Delta} \sigma_z \quad (\text{dropping constant term})$$

Defining $\chi \equiv \frac{g^2}{\Delta}$ as the dispersive shift

$$\tilde{H}_{\text{eff}} = \hbar \left(\frac{\Delta + \chi}{2} \right) \sigma_z + \hbar \chi a^\dagger a \sigma_z$$

Or, returning to the lab frame

$$\hat{H}_{\text{disp}} = \hbar (\omega_n + \chi \hat{\sigma}_z) \hat{a}^\dagger \hat{a} + \frac{\hbar}{2} (\omega_q + \chi) \hat{\sigma}_z$$

Now, we have

$$[H_{\text{disp}}, \sigma_z] = [H_{\text{disp}}, a^\dagger a] = 0.$$

Both the photon number and σ_z are constants of motion and no energy is exchanged. So, a dispersive QND measurement can be performed, i.e., by measuring these observables we do not alter the system

Also, there is a state dependent resonance pull

$$\omega_n^{\text{disp}} \equiv \omega_n \pm \chi \begin{cases} + \text{ when } \sigma_z = +1, \text{ qubit at the excited state} \\ - \text{ when } \sigma_z = -1, \text{ qubit at the ground state} \end{cases}$$

And a qubit frequency modification as

$$\omega_q^{\text{disp}} = \omega_q + \underbrace{\chi}_{\text{Lamb shift}} + \underbrace{2\chi a^\dagger a}_{\text{AC-stark shift (per photon)}}$$